

Velvet Data:

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Assembly      k31_contigs k41_contigs k51_contigs k61_contigs k71_contigs k81_contigs k91_contigs
# contigs (>= 0 bp)      284      207      188      190      197      206      205
# contigs (>= 1000 bp)   218      157      125      110      106      100      106
# contigs (>= 5000 bp)   149      101      88       77       74       73       75
# contigs (>= 10000 bp)  110      89       81       70       69       67       68
# contigs (>= 25000 bp)  53       62       61       54       54       54       55
# contigs (>= 50000 bp)  22       33       34       31       30       31       31
Total length (>= 0 bp)  4730393  4739993  4751867  4760275  4769816  4779309  4782344
Total length (>= 1000 bp) 4701647  4719000  4723421  4726557  4730826  4735667  4739485
Total length (>= 5000 bp) 4537621  4592065  4648496  4655811  4657630  4675555  4668787
Total length (>= 10000 bp) 4247119  4498888  4585046  4599344  4611703  4627574  4615568
Total length (>= 25000 bp) 3268088  4048660  4251032  4341021  4378755  4430345  4417885
Total length (>= 50000 bp) 2151909  2983981  3262090  3522621  3520055  3595432  3537111
# contigs      239      168      144      131      131      128      135
Largest contig 196269    183447    240961    355071    359242    359225    296428
Total length   4716231  4727321  4737219  4741271  4748940  4756430  4761317
GC (%)         50.81    50.81    50.81    50.81    50.81    50.81    50.81
N50            46594    62698    68099    112074    106280    112389    112588
N90            10051    18486    24209    26854    27137    28943    28963
auN            66216.5  77782.6  95499.6  122725.4  127675.1  129200.7  115057.4
L50            27       22       19       14       14       14       15
L90            110      72       62       52       50       49       51
# N's per 100 kbp 32.19    28.77    35.44    22.53    22.34    17.16    2.52
(ecoli_asm) dikonda@login2:~/genomic1/quast_velvet> |

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Oases Data:

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K      N_transcripts  Total_bases  N50_like  Largest_transcript  Dir
31     16511      8620264 516      239620  vel_k31
41     2447       15941700 42133    270152  vel_k41
51     5938       13211021 13387    867110  vel_k51
61     1434       13975317 751529   1204768  vel_k61
71     2772       13613221 55188    721518  vel_k71
81     0          0          0          0        vel_k81
91     0          0          0          0        vel_k91

BEST_OASES
BEST_K 61
DIR     vel_k61
N_transcripts 1434
Total_bases   13975317
N50_like      751529
Largest_transcript 1204768
(ecasm) dikonda@login1:~/Genomics/results/velvet> client_loop: send disconnect: Connection reset

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BEST_VELVET
BEST_K 91
N50    112588
COUNT 230
TOTAL  4785447
LARGEST 366094
DIR     vel_k91
BEST_OASES
BEST_K 61
N50    751529
COUNT 1434
TOTAL  13975317
LARGEST 1204768
DIR     vel_k61

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- Genome assembly was performed using both Velvet and Oases at multiple k-mer values (31, 41, 51, 61, 71, 81, and 91).
- For Velvet, the optimal assembly was achieved at $k = 91$, with an N50 of 112,588 bp, 230 contigs, a total assembly length of 4.79 Mb, and the largest contig measuring 366,094 bp. For Oases, the best performance was obtained at $k = 61$, yielding a much higher N50-like of 751,529 bp, but with 1,434 transcripts and a total base count of 13.98 Mb, with the largest transcript at 1,204,768 bp.
- Although Oases produced longer N50 values, it is designed primarily for transcriptome assembly and generated many more fragments, inflating total assembly length beyond the expected *E. coli* genome size. In contrast, Velvet generated a more accurate, coherent genome assembly closer to the known size (4.6 Mb).
- Therefore, Velvet is the more suitable assembler for this genome dataset.